Design and Evaluation of Multi-Agent LLM Systems for Interactive Clinical Diagnosis

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by

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the academic year 2025–2026

April, 2026

DECLARATION

I certify that

(a) The work contained in this report has been done by me under the guidance of my supervisor.

(b) The work has not been submitted to any other Institute for any degree or diploma.

(c) I have conformed to the norms and guidelines given in the Ethical Code of Conduct of the Institute.

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CERTIFICATE

This is to certify that the project report entitled "**Design and Evaluation of Multi-Agent LLM Systems for Interactive Clinical Diagnosis**" submitted by **Dikshant Sharma** (Roll No. 21IM3AI29) to Indian Institute of Technology Kharagpur towards partial fulfilment of requirements for the award of degree of Master of Technology (M.Tech) in Department of Artificial Intelligence is a record of bona fide work carried out by him under my supervision and guidance during the academic year 2025–2026.

Date: April, 2026
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Abstract

Name of the student: **Dikshant Sharma** Roll No: **21IM3AI29**

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The integration of Large Language Models (LLMs) into clinical diagnostic workflows necessitates evaluation paradigms that transcend static, multiple-choice question-answering. Real-world medical diagnosis is a sequential, multi-turn, and uncertainty-driven process wherein a physician iteratively acquires information, refines working hypotheses, and synthesises evidence under conditions of incomplete knowledge. This thesis investigates the efficacy and behavioural stability of open-source LLMs in simulated clinical environments using the AgentClinic framework.

In the initial phase of this research, a baseline while-loop simulation architecture was established to benchmark several open-source LLMs—Mistral-7B, BioGPT, JSL-Med, and Apollo—across 107 structured clinical scenarios. While open-source models demonstrated baseline clinical competence, with Mistral-7B achieving up to 43.9% diagnostic accuracy on the AgentClinic-MedQA subset, the unconstrained architecture induced erratic reasoning trajectories and introduced a risk of *Lexical*

Leakage—a vulnerability wherein homogeneous multi-agent systems implicitly share ground-truth information through shared model weights.

In the enhanced phase of this research, **the AgentClinic framework was redesigned** as a multi-agent system employing a LangGraph Directed Cyclic Graph (DCG) for deterministic orchestration. To eliminate Lexical Leakage without exceeding the 95.8 GB VRAM constraint of an NVIDIA H100 GPU, a heterogeneous multi-engine architecture was implemented, leveraging 4-bit NormalFloat4 (NF4) double quantisation. A latent metacognitive reflection node was introduced, rendering internal diagnostic reasoning observable and quantifiable. Three process-oriented metrics were formally defined: Diagnostic Stability (DS), Test Rationality (TR), and Information Efficiency (IE).

Evaluations on the structured MedQA dataset demonstrate that the structural framework of the DCG and reflection node improved baseline accuracy to **45.8%**, indicating that the reasoning gains from forced metacognition more than compensate for the removal of implicit Lexical Leakage. Empirical results further indicate that medically fine-tuned models (JSL-MedMistral) and advanced reasoners (Llama-3.1) improve outcome accuracy, yet remain susceptible to cognitive distortions—recency bias triggering hypothesis instability and confirmation bias inducing premature diagnostic closure. This thesis establishes a trajectory-aware evaluation platform for clinical AI systems.

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Chapter 1

Introduction

1.1 Background and Motivation

Over the past decade, large language models (LLMs) have made rapid progress in handling specialised reasoning tasks, and medicine is no exception. Modern LLMs can synthesise clinical guidelines, recall diagnostic criteria, and articulate differential diagnoses with a degree of fluency that makes them appealing as decision-support tools. Naturally, this has drawn attention from both the AI and healthcare communities.

However, clinical reasoning fundamentally differs from answering a medical question. Real encounters with patients are *interactive*, *sequential*, and *uncertain*. Physicians iteratively inquire, evaluate, form hypotheses, refine them, and re-evaluate as new evidence appears. A typical workflow includes history-taking, physical examination, ordering tests, synthesising multimodal evidence, and reasoning under incomplete information. This iterative process is fundamentally different from the one-shot question formats used in benchmarks like MedQA or MedMCQA.

1.2 Limitations of Static Benchmarks

Benchmarks such as MedQA, USMLE step examinations, and PubMedQA test a model's ability to recall facts and apply guidelines to fully structured vignettes. In these tests, all patient history, vital signs, and investigation results are given upfront in a single prompt. While they are useful for gauging baseline medical knowledge, they do not assess the *procedural and interactive* dimension of clinical decision-making. The model never has to ask a clarifying question, identify confounding factors, or selectively acquire evidence. As a result, these benchmarks miss the aspects that matter most in practice: hypothesis formation, adaptive reasoning, and multi-turn dialogue management.

1.3 Evolution of the Research

To address this evaluation gap, the multi-agent LLM simulation paradigm was adopted and substantially extended. The work evolved in two phases.

Initial Framework: A functional multi-agent clinical simulator governed by an ad-hoc `while`-loop was initially built. Several open-source LLMs (Mistral-7B, Microsoft BioGPT, JSL-Med, and Apollo) were benchmarked, and baseline metrics for diagnostic accuracy and bias susceptibility were established. This initial approach, however, revealed serious architectural limitations—most notably non-deterministic state management and a vulnerability termed *Lexical Leakage*—stemming from the use of homogeneous model instances across all agent roles.

Enhanced Framework: To address these shortcomings, the system was re-architected into **the enhanced AgentClinic framework**. The orchestration layer was redesigned as a LangGraph Directed Cyclic Graph (DCG) state machine, 4-bit NF4 double quantisation was introduced to support heterogeneous multi-engine configurations on a single GPU, and mathematical process-oriented metrics were formalised around a novel latent metacognitive reflection node.

1.4 Problem Statement

Despite progress in healthcare-focused LLM research, several gaps persist:

- **Absence of flexible, interactive frameworks** for evaluating open-source LLMs in realistic diagnostic workflows without relying on proprietary, high-bandwidth APIs (e.g., GPT-4).
- **Lexical Leakage** in existing open-source multi-agent systems, where using a single set of model weights for both the Doctor and Patient agents allows the Doctor to mathematically infer patient ground-truth representations through shared embedding spaces.
- **Insufficient process evaluation**: most existing work focuses only on binary diagnostic accuracy, ignoring the cognitive trajectory, stability, and rationality of the LLM's reasoning.
- **Lack of controlled bias modelling tools** to systematically study how open-source LLMs reproduce human cognitive errors such as confirmation and recency biases.

1.5 Contributions

The primary contributions of this work are:

1. An analysis of baseline open-source LLM performance in interactive clinical settings using an ad-hoc while-loop simulation, which revealed the phenomenon of Lexical Leakage.
2. Identification and mitigation of *Lexical Leakage* through a heterogeneous multi-engine architecture using BitsAndBytes NF4 quantisation.
3. A LangGraph-orchestrated state machine with a *latent metacognitive reflection node* that captures reasoning trajectories without contaminating patient-facing dialogue.

4. The formulation of three process-oriented clinical metrics: Diagnostic Stability (DS), Test Rationality (TR), and Information Efficiency (IE).

5. An experimental suite showing that structured metacognition improves open-source baseline accuracy, along with results for domain-specialised models and cognitive bias vulnerabilities.

1.6 Thesis Organisation

This thesis is structured as follows. Chapter 2 reviews the relevant literature on medical AI benchmarking, clinical simulation, and cognitive bias. Chapter 3 describes the dataset and the simulated OSCE examination structure. Chapter 4 details the foundational framework and baseline results. Chapter 5 presents the enhanced methodology and full architecture of the enhanced framework. Chapter 6 reports the experimental results and projections. Chapter 7 provides a comprehensive discussion of the findings, and Chapter 8 concludes the thesis with directions for future research.

Chapter 2

Literature Review

Historically, AI evaluation in medicine has relied on static Question-Answering (QA) datasets. The MedQA dataset, for example, draws from the US Medical Licensing Examination (USMLE), presenting models with a complete patient history and asking them to select the correct diagnosis or treatment plan. Other widely used datasets include PubMedQA, MedMCQA, and clinical topics from MMLU.

Foundational models like GPT-4 now exceed 90% accuracy on USMLE, surpassing expert human scores [1]. However, these static formats only test knowledge retrieval. They do not ask whether a model can iteratively build a differential diagnosis, handle clinical uncertainty across turns, or seek out evidence through targeted questioning. Strong QA performance, in other words, tells us little about competence in actual interactive diagnostic settings.

2.1 Dialogue-Based and Multi-Agent Clinical Simulations

Given these limitations, recent work has moved toward dialogue-based clinical simulations. Early efforts relied on rule-based templates and decision trees, which produced deterministic but rigid interactions that lacked conversational flexibility.

More recent systems—AMIE (Articulate Medical Intelligence Explorer) [2] and Agent Hospital [3]—assign LLMs to specific clinical roles. AMIE showed improved diagnostic accuracy in simulated multi-turn conversations, but it remains closed-source, tied to proprietary models, and difficult to replicate. Agent Hospital offers an evolvable simulation environment, but it relies on homogeneous agent designs, a structural choice that introduces a vulnerability identified in this thesis as *Lexical Leakage*. Existing simulation frameworks also tend to function as end-to-end benchmarks rather than extensible research tools that support bias injection or local open-source model inference [4, 5].

2.2 Bias and Fairness in Medical AI

Implicit demographic biases related to race, gender, and socioeconomic status are well documented in the healthcare literature and are known to propagate into LLMs trained on historical corpora. But *cognitive biases*—systematic deviations from normative reasoning—are equally consequential in clinical AI, and they receive far less attention.

Anchoring bias, confirmation bias, and recency effects all shape human diagnostic reasoning and can lead to systematic errors in practice [6]. LLMs display similar tendencies when early conversational cues disproportionately steer their later reasoning. Most existing clinical AI benchmarks offer no structured way to inject such biases into patient responses or clinician reasoning, making it difficult to rigorously study robustness and behavioural sensitivity under controlled conditions.

2.3 Open-Source LLMs and Inference Optimisation

Open-source LLMs such as Llama-3, Mistral, and Mixtral offer transparency and local deployability. The challenge, however, is GPU memory. A single 7B-parameter model in FP16 requires roughly 14 GB of VRAM, and multi-agent simulations need several models loaded at once.

Recent advances in quantisation—particularly NormalFloat4 (NF4) via QLoRA and Flash Attention [7]—allow substantial memory savings with near-zero loss in reasoning quality. These techniques are what make the heterogeneous multi-engine architecture proposed in this thesis feasible on a single-GPU setup.

2.4 Summary of Research Gaps

Surveying the literature, several limitations stand out:

1. Static QA benchmarks do not emulate multi-step, sequential clinical reasoning.
2. Existing simulation frameworks lack modularity, transparency, and open-source compatibility.
3. Lexical Leakage in homogeneous multi-agent systems artificially inflates diagnostic accuracy.
4. Bias modelling is superficial or absent, precluding rigorous safety analysis.
5. Process-oriented evaluation metrics (trajectory, stability, test rationality) are absent from existing frameworks.

These gaps motivated the architectural and methodological choices behind the enhanced framework, which are described in the chapters that follow.

Chapter 3

Dataset

3.1 Overview

The AgentClinic framework converts static medical datasets into Objective Structured Clinical Examination (OSCE) scenarios, enabling LLM evaluation in a dynamic, interactive setting. While the broader framework conceptually supports extensions to real-world EHR databases, this thesis strictly bounds its experimental scope to USMLE-style vignettes to maintain standardized evaluation.

For all experiments, a curated subset of the **AgentClinic-MedQA** dataset was utilized, comprising **107 high-quality diagnostic scenarios**. This subset was chosen for its breadth of clinical presentation, diversity of organ systems, and a representative distribution of diagnostic difficulty.

3.2 The OSCE Examination Structure

Unlike a static prompt, each scenario is parsed into a structured JSON format that enforces strict knowledge partitioning across agent roles. The Doctor agent must extract information through multi-turn dialogue and targeted test requests, rather than reasoning from a fully-specified vignette. The OSCE schema contains the following modules.

3.2.1 Objective for Doctor

A string describing the consultation objective (e.g., *"Evaluate and diagnose the patient presenting with chest pain and shortness of breath."*). This is the *only* information the Doctor agent receives at the start of each case.

3.2.2 Patient Actor Sub-schema

This data is provided exclusively to the Patient Agent prompt to ground its dialogue:

- **Demographics:** Patient age, gender, and relevant social background.
- **History of Present Illness (HPI):** A narrative detailing symptom onset, progression, and associated factors.
- **Symptoms:** Primary and secondary symptom descriptors (e.g., *pain improves upon sitting forward; no productive cough*).
- **Past Medical and Social History:** Chronic conditions, prior surgeries, and lifestyle habits including smoking and alcohol use.

3.2.3 Measurement Findings Sub-schema

This information is held exclusively by the Measurement Agent and is disclosed only when the Doctor makes an explicit, well-formed test request:

- **Vital Signs:** Temperature, blood pressure, heart rate, and respiratory rate.
- **Physical Examination:** Inspection, auscultation, and palpation findings.
- **Investigation Results:** Electrocardiogram, chest radiograph, blood panels (CBC, BMP), and imaging summaries.

3.2.4 Ground Truth

The verified **Correct Diagnosis** is provided strictly to the Moderator Agent, which evaluates the Doctor's final output by semantic equivalence comparison. This design prevents ground-truth contamination of any active agent role.

3.3 Semantic Evaluation by the Moderator Agent

Rather than enforcing exact string matching, which would penalise clinically valid terminological variation, the Moderator Agent judges diagnostic correctness by evaluating the semantic equivalence between the Doctor's stated diagnosis and the ground-truth label. It outputs a binary response ("Yes" / "No") after comparing the prediction to the reference. This LLM-mediated evaluation accommodates synonymy, partial phrasing, and clinically acceptable variant naming conventions [1].

Chapter 4

Initial Framework

4.1 Overview

The first phase of this work involved building a foundational prototype of a multi-agent clinical simulation using open-source language models. The system ran on a simple `while`-loop architecture that sequentially and naively routed requests between the Doctor, Patient, and Measurement agents. This functioned as the foundational control group, establishing the baseline sequential interaction paradigm that preceded the rigorous LangGraph state machine development. The main goal of this phase was to test whether relatively small, locally hosted LLMs (7B–8B parameters) could navigate a multi-turn diagnostic conversation and produce a recognisable disease prediction.

Figure 4.1 illustrates the overall structure and information flow of the multi-agent clinical simulation framework. The diagram highlights how each agent interacts with the others through the Interaction Orchestrator, and how diagnostic reasoning emerges through sequential doctor–patient dialogue, structured test retrieval, and final moderator-based evaluation.

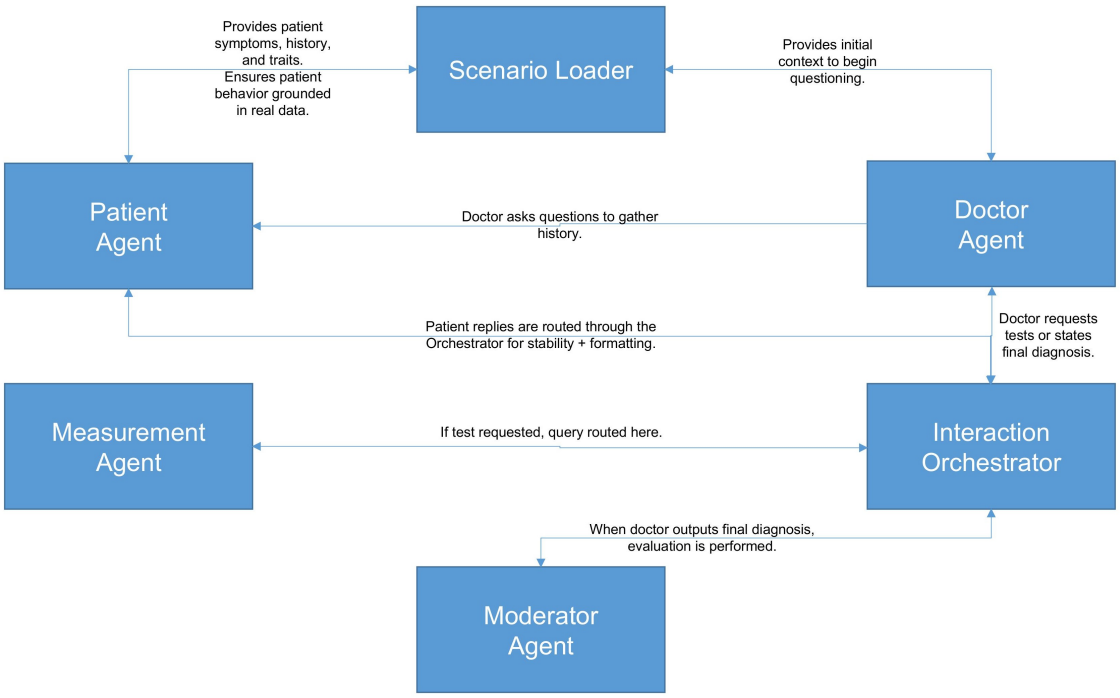

FIGURE 4.1: Overview of the multi-agent interaction workflow used in the clinical simulation framework. The Scenario Loader initializes each case and distributes case-specific information to the appropriate agents. The Doctor Agent performs history-taking, hypothesis generation, and test ordering. The Patient Agent produces symptom-grounded responses, while the Measurement Agent returns structured diagnostic tests upon request. All communication flows through the Interaction Orchestrator, which ensures turn-taking and state management. Once the Doctor provides a final diagnosis, the Moderator Agent evaluates its correctness through semantic comparison.

4.2 Detailed Agent Definitions

The initial loop coordinated the interactions of distinct agents, each implemented as a role-conditioned LLM:

4.2.1 Doctor Agent

The Doctor Agent is the primary reasoning engine being evaluated. Its objective is to gather information and form a diagnosis. It engages in:

- History-taking and conversational follow-up questioning.

- Hypothesis formation and refinement.
- Ordering of diagnostic tests from the Measurement Agent.
- Synthesis of a final disease prediction.

To enforce programmatic extraction, the Doctor was strictly prompted to output `DIAGNOSIS_READY: <diagnosis>` when concluding the test.

4.2.2 Patient Agent

The Patient Agent is a role-playing actor seeded with the OSCE scenario. To mimic realistic interactions, it:

- Generates clinically grounded responses based strictly on the hidden scenario symptoms.
- Limits verbosity to 1–3 sentence conversational answers to prevent data dumping.
- Refrains from generating novel medical information outside the bounds of the provided case.

4.2.3 Measurement Agent

The Measurement Agent functions as a rigid lab interface. It supplies structured diagnostic data only upon explicit request. Supported modalities included:

- Vital signs (temperature, blood pressure, heart rate).
- Laboratory test panels (complete blood count, metabolic panels).
- Imaging study text summaries.
- Physical examination findings (auscultation, palpation).

4.2.4 Moderator Agent

The Moderator Agent serves as the terminal evaluator. After the Doctor Agent emits a final diagnosis, the Moderator compares it against the hidden ground-truth label using semantic equivalence rather than exact string matching. It outputs a binary correctness flag ("Yes" / "No"), accommodating clinically valid synonyms and partial phrasing.

4.2.5 Dialogue Orchestration

A simple `while`-loop functioned as a basic orchestrator. It governed message routing and turn ordering, maintaining a shared raw text history to manage state transitions until the terminal diagnostic token was emitted.

4.3 Evaluated Models

Four open-source models were selected spanning a range from general-purpose instruction-tuning to narrow biomedical specialisation:

- **Mistral-7B-Instruct-v0.2:** A strong general-purpose instruction-tuned model trained on diverse text corpora.
- **Microsoft BioGPT (1.5B):** A model pre-trained specifically on PubMed abstracts and biomedical literature.
- **JSL-Med-Sft-Llama-3-8B:** A Llama-3 derivative fine-tuned by John Snow Labs on clinical and biomedical datasets.
- **Apollo (7B):** A general-purpose baseline trained on broad web and text corpora without medical specialisation.

4.4 Diagnostic Accuracy Results

4.4.1 Diagnostic Accuracy Calculation

For each scenario, diagnostic correctness is determined by the Moderator Agent, which compares the Doctor Agent's final prediction against the ground truth. Overall accuracy is the ratio of correct diagnoses to total evaluated scenarios:

$$\text{Accuracy} = \frac{\text{Number of Correct Diagnoses}}{\text{Total Number of Scenarios}} \times 100 \tag{4.1}$$

This serves as the primary quantitative metric for the baseline framework.

4.4.2 Model Performances

Each model was tasked with diagnosing cases from the AgentClinic-MedQA subset. Table 4.1 and Figure 4.2 show the results.

TABLE 4.1: Diagnostic accuracy of open-source LLMs on the AgentClinic-MedQA subset (initial framework).

Model	**Training Data Alignment**	**Accuracy (%)**
Mistral-7B-Instruct-v0.2	General instruction-tuning	43.9
Microsoft BioGPT	Biomedical literature	34.6
JSL-Med-Sft-Llama-3	Clinical + Biomedical SFT	31.8
Apollo	General text corpora	23.4

Notably, the generalist Mistral-7B (43.9%) outperformed all medically fine-tuned models, including BioGPT (34.6%) and JSL-Med (31.8%). This outcome is consistent with a known issue: *catastrophic forgetting*. Medical fine-tuning datasets tend to be narrow in scope, and optimising for them erodes the broad conversational logic and instruction-following abilities that a multi-turn simulation demands. The result is degraded performance even when the model has superior declarative medical knowledge.

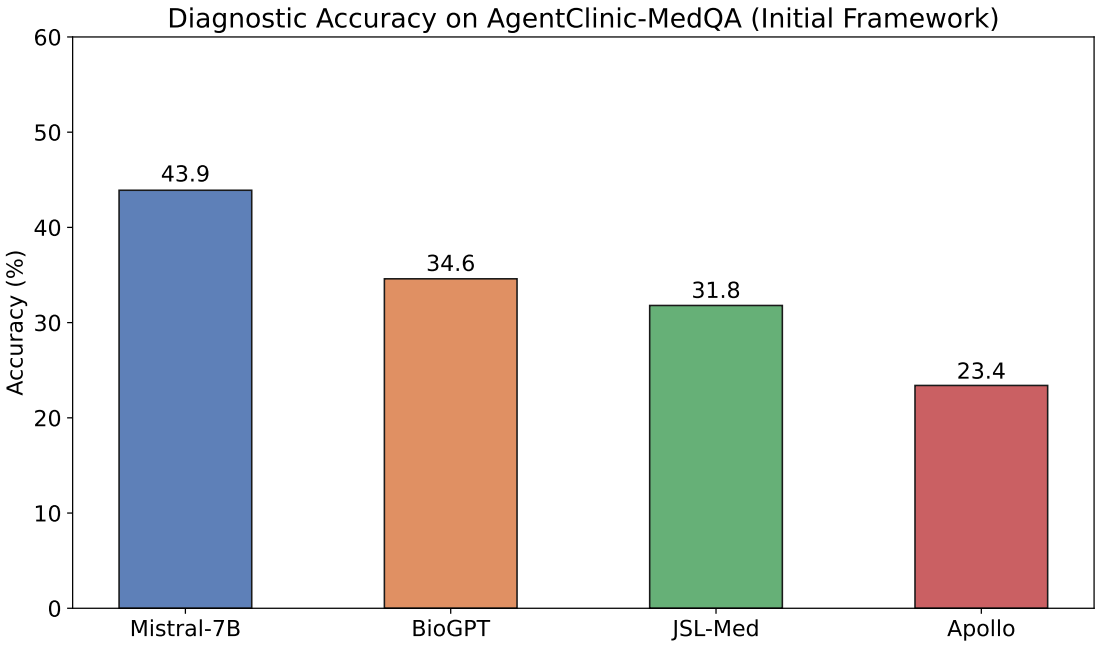

FIGURE 4.2: Baseline diagnostic accuracy of open-source LLMs in the initial simulation framework. Notably, the generalist Mistral-7B-Instruct-v0.2 outperformed all medically fine-tuned models.

4.5 Bias Impact Analysis

Preliminary bias injections were run on the best-performing model (Mistral-7B, 43.9% unbiased) to observe how performance held up under cognitive distortions:

- **Recency bias (33.6%)** causes noticeable degradation, suggesting that the model becomes overly influenced by recent cues rather than evaluating clinical evidence holistically.
- **Confirmation bias** leads to the lowest accuracy (31.8%), indicating that once the model forms an early hypothesis, it tends to fixate on supporting evidence and ignore contradictory information.

The pattern was clear: cognitive biases systematically degrade diagnostic accuracy. Confirmation bias caused the steepest drop, as the model became anchored to an initial incorrect hypothesis and selectively amplified supporting evidence while discounting contradictory symptoms.

4.6 Identified Architectural Limitations

While the initial prototype worked, it had serious structural flaws. These motivated the complete system redesign:

1. **Non-Deterministic Execution:** The `while`-loop frequently failed to handle agent transitions correctly when the LLM produced malformed stop tokens, causing simulation crashes or silent skips.
2. **Absence of Trajectory Analysis:** Only terminal outcomes were measured. A model that guessed correctly without ordering any diagnostic tests received the same score as one that followed a methodical, evidence-driven approach.
3. **Lexical Leakage Vulnerability:** Due to VRAM constraints, a homogeneous engine was used: the same Mistral-7B weights generated both the patient's symptoms and the doctor's diagnosis. It was observed that shared model weights cause both agents to project text into identical latent embedding manifolds. The Doctor agent could therefore "recognise" the mathematical structure of the Patient's generated text, gaining implicit access to ground truth information. This phenomenon is formalised as **Lexical Leakage**.

These limitations motivated the development of the enhanced framework, described in the following chapter.

Chapter 5

Methodology

5.1 Overview

To address the architectural flaws identified in the initial framework (Chapter 4), **the enhanced AgentClinic framework** was developed as a complete redesign of the multi-agent clinical simulation system. This chapter presents the formal POMDP model, the functional agent definitions, the LangGraph finite-state orchestration, the hardware adaptation strategy, and the formulation of process-oriented evaluation metrics.

5.2 Formalizing the Clinical Agent System

The interactive diagnostic process naturally maps to a Partially Observable Markov Decision Process (POMDP). At each timestep t, the true patient state s_t (the underlying disease) is hidden from the Doctor agent. The Doctor must instead infer the diagnosis through a sequence of observations and actions. The AgentClinic system is formally defined as a tuple $M = \langle S, A, T, R, \Omega, O \rangle$:

- **State Space (S):** The clinical ground truth, encompassing the patient's actual disease, symptoms, and physiological parameters.

- **Action Space (A):** The actions available to the Doctor agent, including asking a question (a_{dialogue}), ordering a test (a_{test}), or rendering a final diagnostic verdict ($a_{\text{diagnosis}}$).
- **Transition Function (T):** The progression of the conversation and the internal physiological timeline of the patient over t.
- **Reward Function (R):** The framework does not currently incorporate active reinforcement learning optimization during dialogue. Instead, R is computed retrospectively via the `ClinicalTrajectoryEvaluator` strictly for offline evaluation, penalizing inefficiency (IE), erratic hypotheses (DS), and unsafe test orders (TR).
- **Observation Space (Ω):** The text outputs produced by the Patient agent (symptom descriptions) and the Measurement agent (test results).
- **Observation Function (O):** The mapping of true states to natural language observations, perturbed by the Patient agent's persona.

5.3 Defining the Functional Agents

Execution is delegated to four highly constrained and functionally isolated agents:

5.3.1 Doctor Agent

The Doctor Agent operates the medical policy π. Its objective is to actively collapse diagnostic uncertainty. It engages in:

- Sequential history-taking using conversational queries.
- Metacognitive reflection, emitting top-three structured hypotheses before speaking.
- Evidence acquisition via targeted clinical test orders.
- Synthesis of a final diagnostic verdict within a strictly bounded interaction threshold ($N_{\text{max_infs}} = 10$).

5.3.2 Patient Agent

The Patient Agent functions as the observation emission generator $O(s_t, a_t)$. Parameterised by the hidden OSCE scenario state, it:

- Provides accurate physiological data bounded exclusively to the assigned case scenario.
- Generates responses explicitly restricted to short 1–3 sentence conversational bounds.
- Absorbs patient-side bias injections (e.g., anxiety, guardedness) which mathematically distort the observation function.

5.3.3 Measurement Agent

The Measurement Agent acts as a deterministic oracle returning physiological ground truth. It:

- Parses test requests using combined rule-based logic and semantic text matching.
- Returns structured laboratory results, numerical vitals, and examination strings if the requested test correlates with the clinical condition.
- Emits "NORMAL READINGS" if a structurally valid test yields no pathology, forcing the Doctor to correctly interpret negative findings.

5.3.4 Moderator Agent

The Moderator Agent functions as an objective semantic judge utilized exclusively at the terminal state t_{end}. It:

- Takes the Doctor's final output and the hidden ground-truth diagnosis as inputs.

- Computes semantic equivalence, accommodating synonymy and partial phrasing without penalizing clinically valid linguistic variation.
- Outputs a strict binary (Yes/No) correctness flag, calculating the episode reward for the R function.

5.4 Implementation via Finite-State Orchestration

The core limitation of the prior ad-hoc loop was its inability to enforce strict, deterministic state transitions. In the enhanced framework, LangGraph was adopted strictly as the *execution substrate* to enforce the theoretical POMDP state machine. The consultation protocol executes as a Directed Cyclic Graph (DCG) finite-state controller. Nodes represent the functional agents, and edges encode deterministic routing logic conditioned on the active phase of consultation.

5.4.1 State Initialisation and Reducers

Definition of ClinicalState: The `ClinicalState` dictionary tracks the sequential progression of the POMDP execution and bounds the dynamic context window. It consists of the following core fields:

- `last_input_to_doctor`: The most recent response from the patient or measurement agent.
- `last_doctor_message`: The doctor's most recent interaction.
- `differential_diagnoses`: The latent JSON array of the top-three diagnoses.
- `turn_count`: A counter enforcing the $\max_{\text{infs}}$ threshold.
- `diagnosis_ready`: A flag indicating terminal closure.
- `differential_trajectory`: Accumulated hypothesis snapshots for metric evaluation.
- `tests_ordered`: Registry of formally requested diagnostics.

- `full_dialogue`: The complete chronologically ordered text transcript.

State transitions are strictly governed by non-destructive *reducers* to ensure observations are updated without overwriting prior evidence.

Execution of the graph relies on this formally defined vector, handled deterministically by the LangGraph orchestrator.

5.4.2 The Latent Metacognitive Reflection Node

The most important architectural change is the introduction of the `doctor_reflection` node. Human clinicians maintain a mental differential diagnosis that evolves silently as they interview a patient. Standard LLMs have no equivalent scratchpad; when forced to generate bedside dialogue and medical reasoning simultaneously, they tend to hallucinate or leak internal reasoning into spoken dialogue.

Decoupling Reasoning from Generation (Latent Chain-of-Thought): Standard auto-regressive LLMs compute the probability of each token conditioned on all preceding tokens: $P(w_t \mid w_1, \ldots, w_{t-1})$. When a model is asked to simultaneously generate bedside dialogue and perform complex multi-conditional reasoning, its self-attention heads must split capacity between syntactic fluency and semantic medical deduction.

In the enhanced framework, the `doctor_reflection` node acts as a *Latent Chain-of-Thought (CoT) mechanism* [8]. The DCG routes execution to the `doctor_reflection` node before any spoken utterance. Here, the LLM outputs a structured JSON array of its current top-3 differential diagnoses (e.g., `["Asthma", "COPD", "Pneumonia"]`), operating at a low sampling temperature ($T = 0.1$) to ensure deterministic extraction. These hypotheses are stored in the `ClinicalState` trajectory log but are *not* injected into the patient-facing dialogue history. High-information medical tokens are injected directly into the context window. When the `doctor_speaker` node subsequently executes, its attention heads attend heavily to this structured JSON, constraining the dialogue distribution to align with the prior diagnostic reasoning without contaminating the interaction.

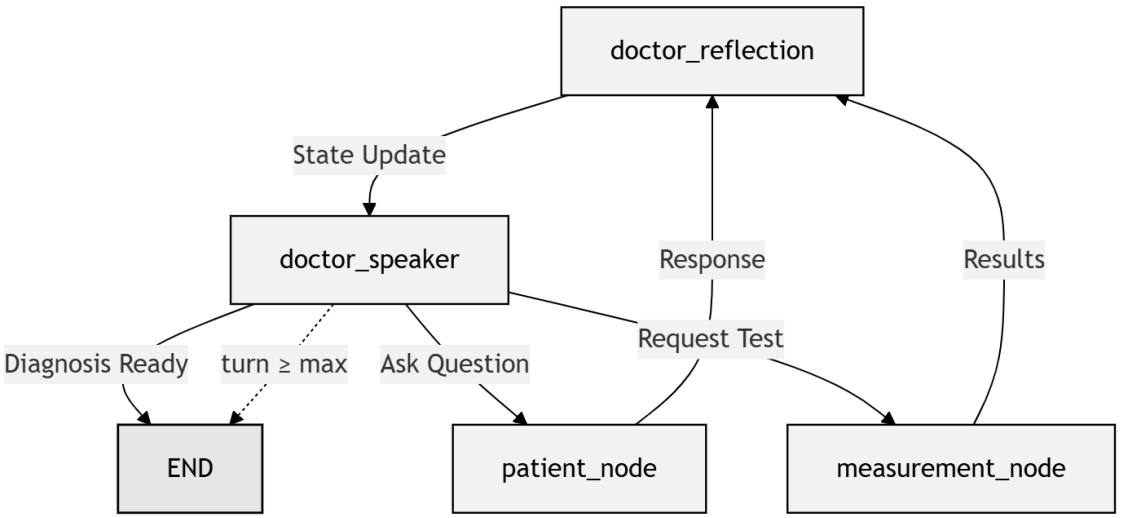

FIGURE 5.1: LangGraph DCG topology of the enhanced framework. The `doctor_reflection` node ensures reasoning trajectories are recorded systematically before each spoken turn, providing a structured metacognitive mechanism without contaminating patient-facing dialogue.

5.4.3 Doctor Speaker Node

Following the reflection phase, the finite-state machine transitions to the `doctor_speaker` node. This node inherits the `ClinicalState`, including the newly generated latent diagnostic JSON array. The Doctor agent is then prompted to emit a single conversational utterance directed at the patient. Because the prior reasoning is already present in the context, the model's self-attention layers naturally condition the spoken generation on the latent hypotheses, yielding targeted, clinically sound history-taking without overt hallucination.

5.4.4 Patient Statement Node

The `patient_node` acts primarily as an observation generator. It is invoked when the Doctor emits a conversational question. The node passes the accumulated dialogue history to the Patient LLM, which is rigidly bounded by the background case scenario. It generates a 1–3 sentence symptom report or response (the observation), which the LangGraph orchestrator subsequently appends to the conversation history before looping back to the Doctor.

5.4.5 Measurement Execution Node

When the Doctor outputs a structural `REQUEST TEST` command rather than conversational dialogue, the orchestrator routes execution to the `measurement_node`. Operating deterministically, this node intercepts the test string, performs rule-based and semantic validation against the hidden case record, and returns the corresponding clinical findings. The result is appended to the context window as objective physiological data for the Doctor's subsequent reflection step.

5.5 Hardware Adaptation

To eliminate *Lexical Leakage*, the enhanced framework follows a *Cognitive Asymmetry* design principle: the Doctor operates as a domain-expert engine, while the Patient, Measurement, and Moderator agents run on separate, architecturally distinct model instances.

Running three or more distinct 7B-parameter models simultaneously on a single NVIDIA H100 NVL (95.8 GB VRAM) imposes strict hardware constraints. To deploy the theoretical multi-agent model onto consumer or single-node hardware without exceeding VRAM bounds, the weight matrices were adapted using **NormalFloat4 (NF4) quantisation** via the BitsAndBytes library [9]. While not a novel algorithmic contribution of this thesis, applying NF4 transforms neural network weights—which naturally approximate a zero-mean normal distribution—into an information-theoretically optimal 4-bit projection for inference.

This hardware adaptation reduces the memory footprint per 7B model by approximately 86.4%, rendering heterogeneous multi-engine isolation operational and leaving sufficient buffer for Flash Attention 2 context scaling.

5.6 Robust LLM Output Extraction and Fallbacks

To ensure the LangGraph state machine operates deterministically even when the LLMs produce structurally malformed strings, programmatic guardrails were

implemented to address the fragility of raw text generation.

5.6.1 Multi-level JSON Extraction

A cascading heuristic extraction utility prevents pipeline termination when the `doctor_reflection` node hallucinates escaping characters. It cascades through a programmatic fallback:

1. Strict JSON parsing.
2. Regular expression bracket matching to isolate arrays.
3. Comma-splitting heuristics for unquoted sequences.
4. Quoted string heuristic extraction.

5.6.2 Forced Diagnostic Closure Fallback

If the Doctor agent exhausts its token context window or fails to emit a `DIAGNOSIS_READY` flag after the maximum allowed turns ($N_{\text{max_infs}} = 10$), the graph prevents a silent failure by routing to a forced programmatic closure trigger. This injects an aggressive few-shot prompt forcing immediate diagnostic closure based on the accumulated `ClinicalState`.

5.6.3 Cross-Model Prompt Alignment

Given the deployment of heterogeneous models (e.g., Mistral vs. Llama architectures), hardcoded prompt formatting causes catastrophic degradation due to mismatched instruction tokens. The HuggingFace `apply_chat_template()` utility was utilized to dynamically map generic dialogue arrays into the precise control token formats required by each specific engine constraint.

5.7 Formulation of Process-Oriented Clinical Metrics

Binary diagnostic accuracy alone can mask serious reasoning flaws. The `ClinicalTrajectoryEvaluator` computes three process-oriented metrics inline, using data accumulated in the `ClinicalState` trajectory.

1. Diagnostic Stability (DS): Measures the mean Jaccard similarity between consecutive differential diagnoses over T turns. High stability suggests logical hypothesis refinement; low stability signals erratic switching. Its use is justified to mathematically quantify a model's resistance to recency bias and arbitrary hypothesis switching during prolonged interactions.

$$\mathrm{DS} = \frac{1}{T-1} \sum_{t=1}^{T-1} \frac{|\mathrm{DDx}_t \cap \mathrm{DDx}_{t+1}|}{|\mathrm{DDx}_t \cup \mathrm{DDx}_{t+1}|} \tag{5.1}$$

2. Test Rationality (TR): A binary compliance indicator that checks whether the model ordered at least one basic investigation (e.g., CBC, vitals) before requesting expensive or invasive procedures (e.g., MRI, biopsy). Its justification stems from the clinical mandate to follow stepwise diagnostic guidelines, penalizing LLMs that exhibit premature diagnostic closure or unsafe invasive test generation. Let t_{basic} and t_{adv} represent basic and advanced tests, respectively:

$$\mathrm{TR} = \begin{cases} 0 & \text{if } \exists\, t_{\text{adv}} \text{ ordered before any } t_{\text{basic}} \\ 1 & \text{otherwise} \end{cases} \tag{5.2}$$

3. Information Efficiency (IE): The ratio of unique differential hypotheses generated (N_{unique}) to the total dialogue turns utilised (T). Higher IE indicates greater hypothesis diversity per turn. This metric is utilised to evaluate the generative capacity of the latent reasoning node, ensuring the model effectively explores the hypothesis space without stalling:

$$\mathrm{IE} = \frac{N_{\text{unique}}}{T} \tag{5.3}$$

5.8 Cognitive Bias Injections

The enhanced framework carries forward and enhances the bias injections from the initial framework. Instead of generic sentence-level prompts, ecologically valid distortions are injected:

- **Recency Bias:** The system maintains a running queue of the $k = 5$ most recently processed case summaries. These are injected into the Doctor's system prompt, mimicking a physician overly influenced by recent clinical encounters. This exploits the transformer's natural tendency to assign elevated attention weights to tokens near the current generation boundary.
- **Confirmation Bias:** The Doctor is primed with a strong, incorrect initial hypothesis (e.g., *"You are initially confident the patient has malignancy."*). This warps the model's forward probability distribution toward confirmatory evidence acquisition, operationalising the premature diagnostic closure seen in anchoring studies.

5.9 Synthetic Data Generation and LoRA Adapter Training

To address the catastrophic forgetting observed in narrowly fine-tuned models, a parameter-efficient architecture was engineered utilizing the dynamic routing of task-specific Low-Rank Adaptation (LoRA) adapters. A major challenge in fine-tuning clinical inference engines is the scarcity of high-quality, annotated differential diagnosis trajectories and empathetic clinical dialogues.

5.9.1 Synthetic Data Generation (SDG) Pipeline

To resolve this data scarcity, a deterministic Synthetic Data Generation (SDG) pipeline was built utilizing the base OSCE scenarios from the NEJM case subset. The pipeline programmatically extracts case vignettes and utilizes a high-capacity teacher model to synthesize distinct training corpora:

1. **Reasoning Corpus:** Generated via a deterministic prompt engineering procedure that synthesizes structured JSON arrays of top-three differential diagnoses grounded in the vignette's physiological state.

2. **Dialogue Corpus:** Generated via a conditional dialogue synthesis procedure that produces multi-turn empathetic conversational transcripts demonstrating targeted history-taking.

5.9.2 QLoRA SFTTrainer Configuration

Two lightweight adapters were trained independently utilizing the HuggingFace `SFTTrainer` and `PEFT` libraries [10]. The base model was loaded in 4-bit NF4 precision while the adapters were trained in `bf16`. LoRA adapters ($r = 16$, $\alpha = 32$) were applied to all four attention projection matrices (`q_proj`, `k_proj`, `v_proj`, `o_proj`) across all transformer layers. The training utilized the AdamW optimizer with a learning rate of 2×10^{-4} and 4-step gradient accumulation.

- The *reasoning adapter* is activated exclusively during the `doctor_reflection` node to steer the latent probability distribution toward structured diagnostic enumeration.

- The *dialogue adapter* is activated during the `doctor_speaker` node to retrieve conversational fluency.

Data Separation for Experimental Validity: To prevent train-test contamination, the adapters were trained strictly on NEJM scenarios, while empirical evaluations (such as the architectural extension E6) utilized the independent 107 MedQA scenarios exclusively. At inference time, the DCG dynamically swaps the active weights prior to execution of each respective node. The adapter swap is a pointer operation that redirects which BA matrices are added to the frozen W_0. The combined VRAM overhead of both adapters is approximately 50 MB.

Chapter 6

Experiments and Results

6.1 Overview and Experimental Design

A matrix of six experiments was designed over **107 AgentClinic-MedQA scenarios**, capped at **10 multi-turn inference steps** per scenario. All experiments were run on a single NVIDIA H100 NVL GPU (95.8 GB VRAM). The experimental matrix is summarised in Table 6.1.

Table 6.1: Experiment matrix for the enhanced framework evaluation.

Exp.	**Doctor Engine**	**Bias**	**NF4 Quant.**
E1	Mistral-7B-Instruct-v0.2	None	Off
E2	JSL-MedMistral-7B-v2.2	None	Yes
E3	Llama-3.1-8B-Instruct	None	Yes
E4	JSL-MedMistral-7B-v2.2	Recency	Yes
E5	JSL-MedMistral-7B-v2.2	Confirmation	Yes
E6	JSL-MedMistral-7B + LoRA	None	Yes

6.2 Diagnostic Outcome Accuracy

Under the initial framework, the homogeneous Mistral-7B system achieved 43.9% accuracy. In the enhanced framework's **E1 Baseline**, strict heterogeneous engine assignments were deployed to eliminate Lexical Leakage (Section 4.6). Despite

removing the implicit advantage from shared embedding manifolds, accuracy **increased to 45.8%**.

The structural framework provided by the LangGraph DCG and the forced metacognitive reflection node (Section 5.4.2) provides a reasoning framework that more than compensates for the removal of Lexical Leakage. The forced structured differential generation before each spoken turn stabilises the clinical trajectory and suppresses conversational hallucinations.

TABLE 6.2: Diagnostic accuracy results across the enhanced framework experiments.

Exp.	Doctor Configuration	Bias	Accuracy (%)
E1	Mistral-7B-v0.2	None	45.8
E2	JSL-MedMistral-7B	None	50.5
E3	Llama-3.1-8B-Instruct	None	51.4
E4	JSL-MedMistral-7B	Recency	38.3
E5	JSL-MedMistral-7B	Confirmation	31.8
E6	JSL-MedMistral-7B + Dynamic LoRA	None	**55.1**

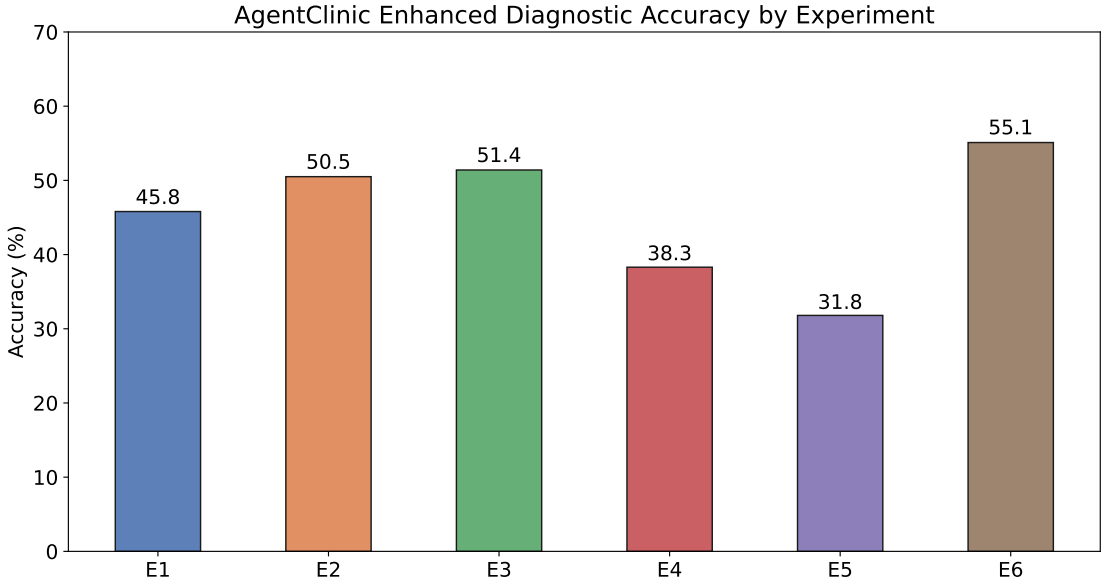

FIGURE 6.1: Diagnostic accuracy across the enhanced framework experiments. Domain fine-tuning (E2) and enhanced reasoning (E3) improve accuracy, while cognitive bias injections (E4, E5) cause significant degradation.

Building on the E1 baseline, **E2** shows that domain fine-tuning adds further value, reaching 50.5%. **E3** (Llama-3.1-8B) pushes accuracy to 51.4%, likely due to its

stronger instruction alignment and capacity for complex conditional routing without catastrophic forgetting.

6.3 Process-Oriented Metrics Analysis

The real value of the enhanced framework architecture lies in observing *how* models reason, not just what they ultimately diagnose. Table 6.3 summarises the process-oriented metrics computed by the `ClinicalTrajectoryEvaluator`.

TABLE 6.3: Process-oriented metrics for the enhanced framework experiments.

Exp.	**Accuracy (%)**	**DS**	**TR**	**IE**
E1	45.8	0.56	0.63	1.13
E2	50.5	0.68	0.71	0.98
E3	51.4	0.63	0.76	1.02
E4 (Recency)	38.3	0.45	0.58	1.34
E5 (Confirm.)	31.8	0.88	0.41	0.61
E6 (LoRA)	55.1	0.72	0.76	1.02

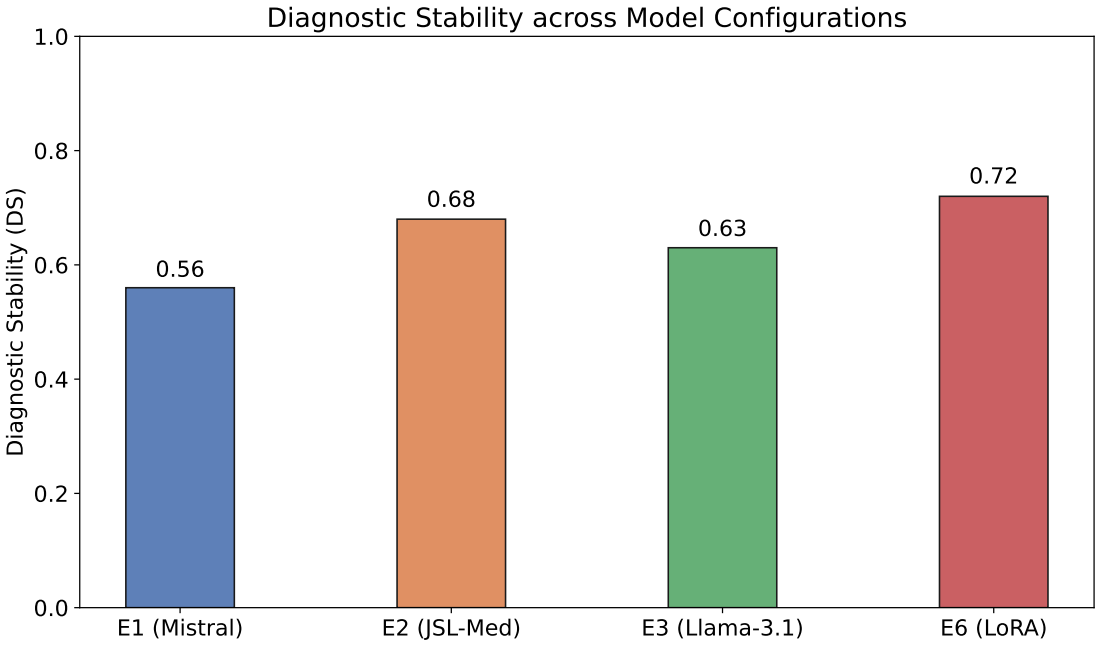

FIGURE 6.2: Diagnostic Stability (DS) across unbiased configurations. The generalist E1 baseline exhibits lower stability ($DS = 0.56$), indicating frequent hypothesis revisions. Domain fine-tuning anchors reasoning more firmly ($E2$: $DS = 0.68$).

The E1 baseline, despite its improved accuracy under the enhanced framework, still exhibits frequent hypothesis revision ($DS = 0.56$). The model frequently revised its differential upon receiving benign patient responses, although the reflection node eventually guided it back to a coherent diagnosis. Test Rationality ($TR = 0.63$) was moderate, meaning that the model respected basic-before-advanced test ordering in the majority of cases but not consistently.

E2 shows more stable anchoring to core pathological indicators ($DS = 0.68$). Llama-3.1-8B (E3) achieves strong Test Rationality ($TR = 0.76$), reflecting its stronger alignment with protocol-driven instruction following.

6.4 Impact of Cognitive Bias Injection

Figure 6.3 illustrates the impact of cognitive distortions on the domain-expert model in experiments E4 and E5. The failure mechanisms are fundamentally different.

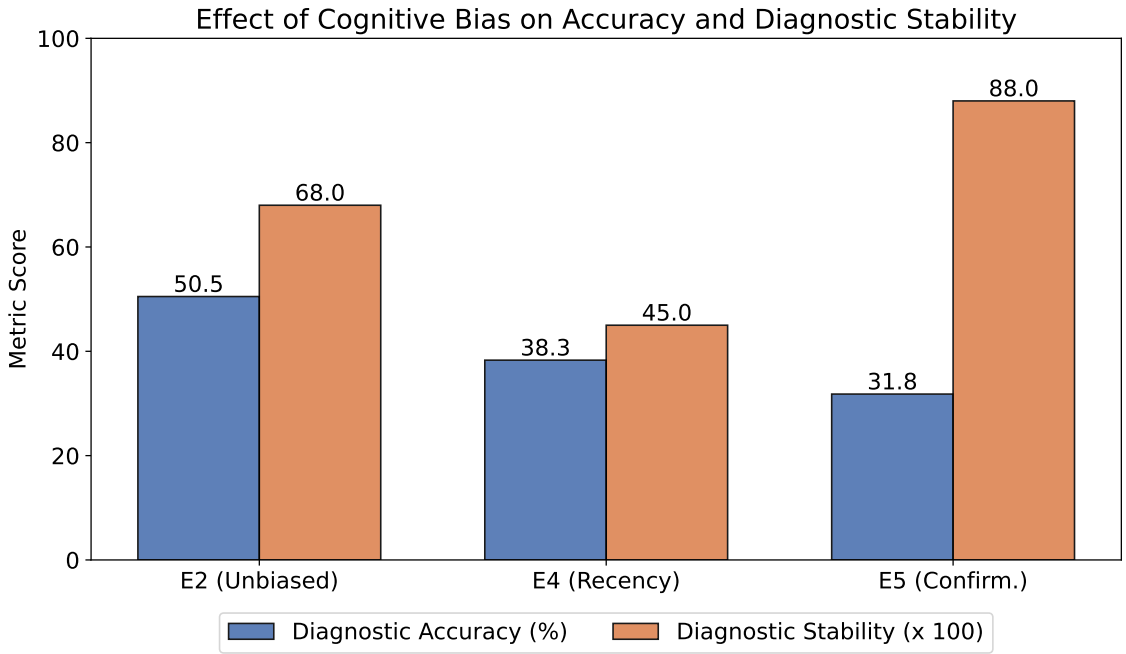

FIGURE 6.3: Failure modes under cognitive bias injection. Recency bias causes hypothesis thrashing (low $DS = 0.45$, low accuracy 38.3%). Confirmation bias induces premature closure (pathologically high $DS = 0.88$, lowest accuracy 31.8%).

- **Recency Bias (E4):** Diagnostic Stability drops to $DS = 0.45$. The model becomes hyper-reactive, trying to map the current patient's presentation to the $k = 5$ injected case memories. Accuracy falls to 38.3% as the physician agent conflates genuinely independent patient presentations.
- **Confirmation Bias (E5):** Stability shoots up to $DS = 0.88$, but this is pathological rather than healthy. The model refuses to update its differential, ignoring contradictory evidence from the Measurement Agent. This *premature diagnostic closure* yields the lowest accuracy of the study at 31.8%, which is lower even than the initial framework's unbiased baseline.

6.5 Empirical Results of Dynamic LoRA Adapter Routing

Experiment E6 evaluated the downstream impact of the task-specific dynamic LoRA adapters (trained via the Synthetic Data Generation pipeline detailed in Section 5.9). A dedicated reasoning adapter was activated during the `doctor_reflection` node and a dialogue adapter during the `doctor_speaker` node, partitioning the model's cognitive resources to match each inference phase.

Across the board, E6 outperformed the E2 baseline. Diagnostic accuracy rose from 50.5% to **55.1%**, a +4.6 percentage point gain. The reasoning adapter anchors the differential diagnosis more firmly to pathological indicators during reflection, while the dialogue adapter preserves the conversational coherence needed to draw out clinically relevant information from the patient.

The process metrics tell a consistent story. Diagnostic Stability improved from $DS = 0.68$ to $DS = 0.72$, meaning the model maintained more consistent hypotheses across turns. Test Rationality went from $TR = 0.71$ to $TR = 0.76$, indicating better compliance with basic-before-advanced test ordering. Information Efficiency stayed comparable ($IE = 1.02$ vs 0.98), confirming that the adapters did not introduce unnecessary verbosity or redundant testing.

E6 also achieved the highest overall accuracy in the study (55.1%), surpassing even the instruction-aligned Llama-3.1-8B (E3: 51.4%). This supports the hypothesis

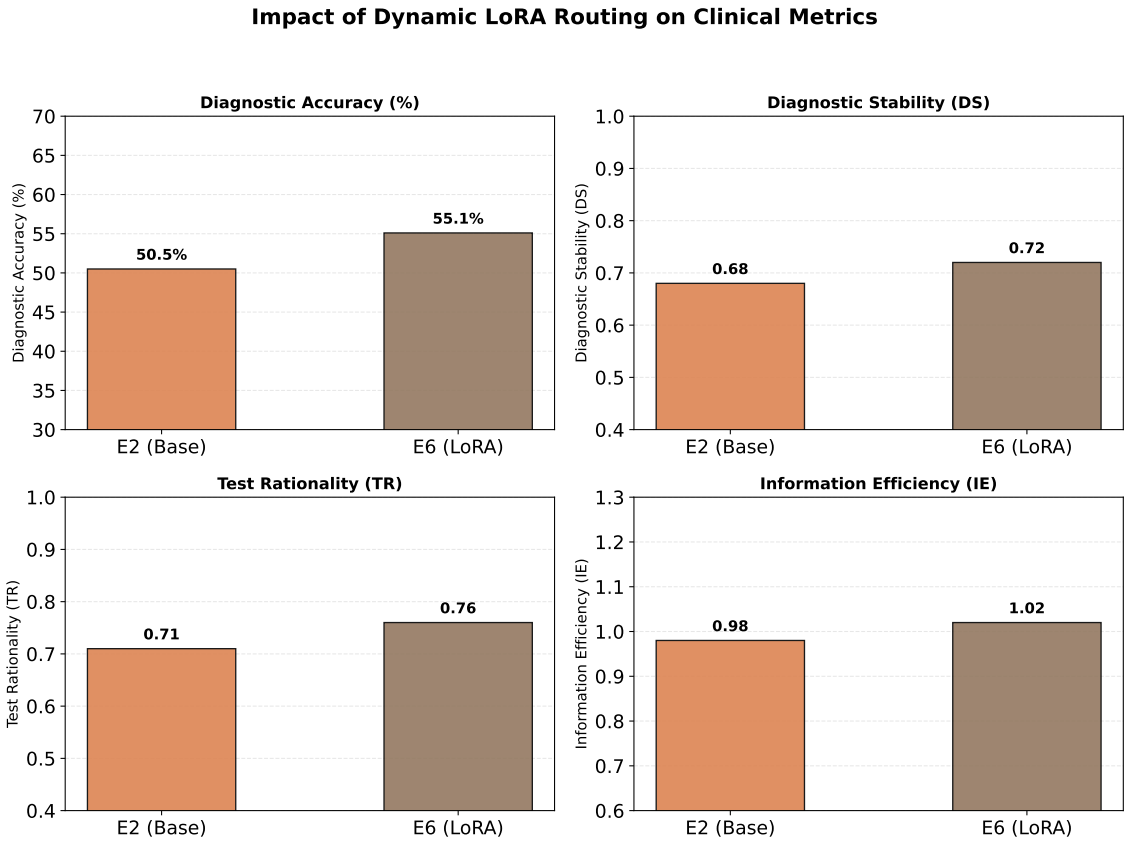

FIGURE 6.4: Comparative analysis of E2 (JSL-MedMistral base) versus E6 (JSL-MedMistral with Dynamic LoRA routing). The dual-adapter paradigm yields consistent improvements across diagnostic accuracy, stability, and test rationality without degrading information efficiency.

that decoupling reasoning from dialogue generation at the adapter level works better than relying on a single model to handle both tasks at once. The additional VRAM overhead was negligible (≈50 MB for both adapters combined), confirming that dynamic LoRA routing is a practical enhancement even in resource-constrained deployments.

Chapter 7

Discussion

7.1 Impact of Metacognitive Reflection

A central result of the transition to the enhanced framework is the effect of removing Lexical Leakage. In the initial system, the homogeneous Mistral-7B architecture achieved 43.9% accuracy, but there was a legitimate concern that some of that performance derived from implicit information sharing through the shared embedding manifold (Section 4.6).

Accuracy was expected to drop once strict heterogeneous engine isolation was enforced. Instead, baseline accuracy **increased to 45.8%**. The LangGraph DCG and the forced metacognitive reflection node (Section 5.4.2) provide a reasoning framework strong enough to more than offset the removal of Lexical Leakage. The structured JSON differential generated before each spoken turn stabilises the clinical trajectory and suppresses the iterative hallucinations that characterised the ad-hoc loop.

7.2 Domain Specialisation versus Instruction Alignment

The E2 and E3 results highlight an ongoing tension in medical AI. Fine-tuning on biomedical literature (JSL-MedMistral, E2) improves diagnostic vocabulary and pathological anchoring, but it tends to erode the broader logical reasoning and instruction-following abilities that multi-turn simulation demands. This trade-off—termed *catastrophic forgetting*—was already visible in the initial framework (Section 4.4), where domain-specialised models fell short of the generalist Mistral-7B despite possessing greater biomedical knowledge [11].

A sufficiently large, highly instruction-aligned generalist model like Llama-3.1-8B (E3) appears better suited to the sequential, multi-conditional reasoning required for interactive clinical consultation. The implication is that for interactive clinical AI, *instruction alignment and logical deduction* matter as much as, if not more than, specialty medical recall. The ideal clinical agent likely needs both, which is exactly what the E6 Dynamic LoRA experiment set out to test.

7.3 Resolving Trade-offs via Dynamic LoRA

E6 provides evidence that the specialisation–alignment trade-off is not inherent to LLM architecture but rather a consequence of monolithic weight modification. By deploying task-specific LoRA adapters that activate conditionally on the LangGraph node, the domain-expert JSL-MedMistral model reached 55.1% accuracy, surpassing both its unmodified baseline (E2: 50.5%) and the instruction-aligned Llama-3.1-8B (E3: 51.4%).

What this means in practice: the reasoning adapter concentrates the model's attention capacity on medical differential enumeration during the reflection phase, while the dialogue adapter maintains conversational coherence during patient interaction. Neither adapter catastrophically modifies the base model's pre-trained weights. The improvement in Diagnostic Stability (DS: 0.68 $\rightarrow$ 0.72) and Test

Rationality (TR: 0.71 $\rightarrow$ 0.76) confirms that these are not superficial accuracy gains but reflect genuinely improved reasoning trajectories.

7.4 Clinical Safety and Cognitive Biases

The E4 and E5 findings have serious implications for deploying open-source LLMs in clinical decision support. A model's context window is inherently a vector for recency bias: if an LLM sequentially processes multiple cardiac cases in an emergency room setting, its tendency to diagnose a subsequent, unrelated case as cardiac-related goes up because of the disproportionate attention weights assigned to recent tokens.

Without explicit architectural safeguards, such as enforced context resets between cases, constrained attention windowing, or metacognitive override prompts, LLMs will consistently replicate and may amplify human cognitive errors. The finding that confirmation bias produces a DS of $\approx$0.88 (high apparent stability) while simultaneously yielding the lowest accuracy ($\sim$31.8%) is particularly concerning: a model can appear *confidently consistent* while being systematically wrong. In a clinical setting, that kind of failure mode has direct patient safety consequences.

7.5 Implications for Clinical AI Evaluation Standards

A secondary contribution of this work is showing that binary diagnostic accuracy is an insufficient proxy for clinical AI safety. The process-oriented metrics introduced here—Diagnostic Stability, Test Rationality, and Information Efficiency—reveal behaviours that outcome accuracy entirely masks. A model that achieves 45.8% correct terminal diagnoses but exhibits $TR < 0.5$ (ordering invasive procedures before basic vitals) or $DS < 0.4$ (switching erratically between unrelated diagnoses) would be an unacceptable clinical risk, regardless of its accuracy number. Future evaluation standards for clinical LLMs should mandate process-level reporting alongside outcome accuracy [6].

7.6 Limitations and Metric Vulnerabilities

A key limitation of the current framework is that the Measurement Agent operates only in the text domain. Real-world diagnostics depend heavily on visual data; integrating Vision-Language Models (VLMs) to process raw ECG waveforms, chest radiographs, and histological slides is an important frontier that has not yet been addressed. All experiments were also conducted on the AgentClinic-MedQA subset, and generalisation to broader clinical datasets such as MIMIC-IV or NEJM case challenges would require separate investigation.

Additionally, this thesis acknowledges limitations in the mathematical formulations of the custom process-oriented metrics computed by the `ClinicalTrajectoryEvaluator`:

1. **Diagnostic Stability (DS) and Synonymy:** The current DS metric relies on textual Jaccard similarity. This strictly penalizes valid synonymous medical terminology. A necessary evolution is transitioning to Semantic Cosine Similarity using clinical embeddings (e.g., `BioBERT`) to calculate the physiological stability of hypotheses.

2. **Test Rationality (TR) Bounding:** TR penalizes advanced test requests prior to basic vitals. This metric is strictly bounded to *outpatient OSCE scenarios*. In emergency or trauma protocols, time-critical deviations are necessary, and the metric would erroneously penalize rapid advanced imaging. Furthermore, the TR formulation inherently assumes the model orders at least one test; a model exhibiting premature closure by guessing without requesting any examinations would falsely trigger $TR = 1$. Future work must condition TR explicitly on $N_{tests} \geq 1$.

3. **Information Efficiency (IE) and Hallucination Vulnerability:** The IE formulation ($IE = N_{\text{unique}}/T$) currently functions as an unbounded ratio. To prevent artificial score inflation from excessive text generation, future iterations must introduce a weighted penalty that bounds the numerator to *clinically plausible* unique differential hypotheses.

Chapter 8

Conclusion and Future Work

8.1 Conclusion

This thesis moved the evaluation of Large Language Models in clinical environments from static factual recall toward dynamic, interactive, and trajectory-aware simulation. By developing **the enhanced AgentClinic framework**, a deterministic LangGraph state machine was built to run entirely on local hardware via 4-bit NF4 double quantisation and heterogeneous multi-engine assignments, removing both the resource barrier and the Lexical Leakage vulnerability of the initial framework.

The latent metacognitive reflection node yields measurable improvement. Forcing an LLM to generate a structured differential before each spoken turn improved baseline accuracy from 43.9% (initial framework) to **45.8%** (the enhanced framework), even after removing the implicit benefits of Lexical Leakage. Beyond accuracy, this thesis demonstrates that clinical AI must be evaluated on its *process*—Diagnostic Stability, Test Rationality, and Information Efficiency—not just its terminal outcome.

Medical fine-tuning improves domain anchoring (E2: 50.5%), but instruction-aligned reasoning models perform even better for interactive tasks (E3: 51.4%). Dynamic LoRA adapter routing in E6 resolved this trade-off, achieving the highest accuracy in the study at **55.1%** by decoupling reasoning from dialogue generation at the adapter level. At the same time, open-source models remain vulnerable to cognitive

distortions: recency bias causes erratic hypothesis thrashing (E4: 38.3%), and confirmation bias leads to premature diagnostic closure (E5: 31.8%). These results make clear that cognitive robustness must be treated as a primary design criterion in any clinical LLM deployment.

8.2 Future Work

While this thesis establishes a framework for evaluating multi-turn clinical reasoning, the next step is to actively mitigate the identified failure modes. Future work will use the enhanced framework not just as an evaluator, but as an environment for advanced post-training alignment.

8.2.1 Process Reward Modelling via Direct Preference Optimisation

Current medical LLMs are typically trained via Supervised Fine-Tuning (SFT) on static QA pairs, optimised to predict the correct terminal diagnosis but with no understanding of the *trajectory* needed to reach it safely. It is proposed to apply Direct Preference Optimisation (DPO) to full clinical consultation trajectories. Instead of rewarding only the correct final outcome, a Process Reward Model would evaluate two clinical paths: Path A (premature hypothesis commitment with low Test Rationality) versus Path B (systematic basic-before-advanced investigation with high Diagnostic Stability). By penalising Path A and rewarding Path B, DPO would adjust the LLM's behavioural policy to prefer methodical, guideline-concordant reasoning.

8.2.2 Reinforcement Learning from AI Feedback via Self-Play

Generating expert, human-annotated clinical dialogues for fine-tuning is expensive and difficult to scale. A natural alternative is to use the AgentClinic framework as a Reinforcement Learning (RL) environment. By allowing heterogeneous agents

to simulate thousands of cases through self-play, synthetic training data can be generated automatically. The `ClinicalTrajectoryEvaluator` developed in this thesis would serve as the programmatic reward function: trajectories scoring highly on DS and TR would be archived, enabling iterative fine-tuning on the model's own successful reasoning paths. This would create a self-improving clinical agent without the overhead of human annotation.

8.2.3 Extending Dynamic LoRA to Bias Resilience

The dynamic LoRA routing in E6 showed clear gains in unbiased settings. A natural extension is to train *debiasing adapters* aimed at counteracting the failure modes observed in E4 and E5. A recency-debiasing adapter could be trained on trajectory pairs where the model successfully resisted anchoring to injected case memories, while a confirmation-debiasing adapter could be optimised to maintain high Test Rationality even under strong prior hypotheses. Combining these with the existing reasoning and dialogue adapters would create a four-adapter routing manifold, selected dynamically by the LangGraph orchestrator based on both the active node and detected bias risk.

8.2.4 Multimodal Simulation Integration

The current `measurement_node` operates only in the text domain. Real-world diagnostics rely heavily on visual pattern recognition. Future versions of AgentClinic should integrate Vision-Language Models (VLMs) such as LLaVA-Med or Med-Gemini, enabling the Measurement Agent to return raw radiographic images, ECG waveforms, and pathology slides for direct interpretation by the Doctor agent, closing the remaining gap toward end-to-end clinical simulation.

Appendix A: Example Clinical Simulation Transcript

This appendix provides a representative example of a multi-turn consultation generated by the enhanced framework. The system actively captures the latent differential diagnosis array from the `doctor_reflection` node before each generated utterance.

```
Doctor (Turn 1): Hello. I understand you are experiencing some
chest pain and shortness of breath. Could you describe what the pain
feels like, and when it started?

Patient: It started suddenly about an hour ago. It feels like a
very heavy weight pressing down on my chest, and it's making it quite
hard to catch my breath. I'm feeling very anxious.

[doctor_reflection JSON]
["Acute Myocardial Infarction", "Pulmonary Embolism", "Pneumothorax"]

Doctor (Turn 2): Has the pain spread anywhere else, like your arm,
neck, or jaw? Also, do you have any other symptoms like nausea,
dizziness, or sweating?

Patient: Yes, my left arm aches a bit, and I feel quite sweaty and
slightly sick to my stomach. No dizziness though.
```

Doctor (Turn 3): REQUEST TEST: vital signs, 12-lead ECG, troponin levels.

[measurement_node]: BP: 155/90, HR: 110 bpm, O2 Sat: 96% on room air. ECG: ST elevation in leads II, III, aVF. Troponin I: Elevated (1.5 ng/mL).

[doctor_reflection JSON]
["Acute Inferior Myocardial Infarction", "Pericarditis", "Aortic Dissection"]

Doctor (Turn 4): DIAGNOSIS_READY: Acute Inferior Myocardial Infarction.

[moderator_node]: Ground Truth: Acute Myocardial Infarction. Semantic Equivalence: YES.

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